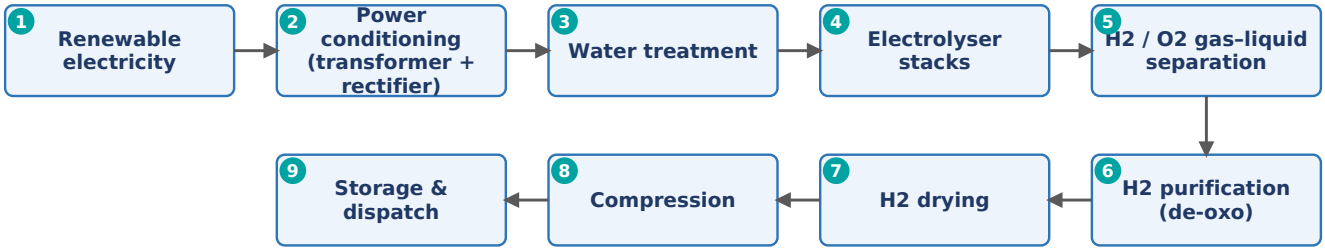
MODULE 3

# Detailed Production Process

Step-by-step: green H2 by electrolysis · grey/blue H2 by SMR

## 3.1 Green hydrogen by water electrolysis

**Process chain – each numbered block is explained below**




### Step 1 – Renewable electricity

| Aspect | Details |
|---|---|
| **Purpose** | Supply low-carbon power; the largest cost item of green $H_2$ [R1] |
| **Inputs** | Solar / wind / hydro power, grid backup or PPA |
| **Outputs** | AC power to plant switchyard |
| **Equipment** | Solar/wind plant, HV line, substation, battery (optional) |
| **Operating parameters** | Capacity factor sets electrolyser operating hours |
| **Process control** | Energy management system, forecasting, power-purchase scheduling |
| **Common problems** | Intermittency lowers utilisation and raises cost per kg |
| **Safety** | HV electrical safety, arc-flash protection |

### Step 2 – Power conditioning

| Aspect | Details |
|---|---|
| **Purpose** | Convert AC to the low-voltage, high-current DC the stacks need |
| **Inputs** | AC from grid/renewables |
| **Outputs** | Controlled DC to each stack |
| **Equipment** | Transformer, thyristor or IGBT rectifier, harmonic filters, DC busbars |
| **Operating parameters** | Stack current set-point follows power available; ripple and power factor per OEM |
| **Process control** | Current/voltage control per stack, power-factor correction |
| **Common problems** | Harmonics, rectifier losses, heat in busbars |
| **Safety** | High DC currents – insulation monitoring, lock-out/tag-out |

## Step 3 – Water treatment

| Aspect | Details |
|---|---|
| **Purpose** | Produce high-purity feed water; impurities poison catalysts and membranes |
| **Inputs** | Raw water (municipal, ground, treated waste or desalinated sea water) |
| **Outputs** | Demineralised water; reject brine |
| **Equipment** | Filters, softener, reverse osmosis (RO), electro-deionisation (EDI) / mixed-bed polishers |
| **Operating parameters** | Water quality to stack OEM specification (conductivity); 18–24 kg raw water per kg $H_2$ incl. treatment losses [R1] |
| **Process control** | Conductivity/TOC analysers, automatic regeneration |
| **Common problems** | Membrane fouling, scaling, silica breakthrough |
| **Safety** | Chemical handling (acids, caustic, antiscalant) |

## Step 4 – Electrolyser stacks

| Aspect | Details |
|---|---|
| **Purpose** | Split water into $H_2$ (cathode) and $O_2$ (anode) |
| **Inputs** | Water (or KOH solution), DC power |
| **Outputs** | Wet $H_2$, wet $O_2$, heat |
| **Equipment** | Cell stacks, end plates, manifolds, circulation pumps, heat exchangers (Module 4) |
| **Operating parameters** | AEL 70–90 °C, 1–30 bar; PEM 50–80 °C, <70 bar; SOEC 700–850 °C [R1] |
| **Process control** | Cell-voltage monitoring, temperature, pressure balance ($H_2$ vs $O_2$ side), $H_2$-in-$O_2$ analyser |
| **Common problems** | Degradation (voltage rise), gas crossover at low load, membrane thinning, catalyst loss |
| **Safety** | $H_2$-in-$O_2$ must stay well below the lower explosive limit; trip on high crossover |

## Step 5 – Gas–liquid separation

| Aspect | Details |
|---|---|
| **Purpose** | Remove entrained water/electrolyte from the gas and return liquid to the stacks |
| **Inputs** | Two-phase gas/liquid from stacks |
| **Outputs** | Gas to purification; liquid back to circulation |
| **Equipment** | Separator drums ($H_2$ and $O_2$ side), demisters, level control valves |
| **Operating parameters** | Level and pressure held equal on both sides (critical for alkaline) |
| **Process control** | Level control, differential-pressure control |
| **Common problems** | Level imbalance can push gas across the separator; electrolyte carry-over |
| **Safety** | Overpressure protection, relief valves to safe vent |

## Step 6 – Hydrogen purification (de-oxo)

| Aspect | Details |
|---|---|
| **Purpose** | Remove trace oxygen that crossed over |
| **Inputs** | Wet $H_2$ with trace $O_2$ |
| **Outputs** | $H_2$ with $O_2$ removed (+ a little water) |
| **Equipment** | Catalytic deoxidiser (precious-metal catalyst) – $O_2$ reacts with $H_2$ to form water |
| **Operating parameters** | Inlet $O_2$ content, catalyst temperature |
| **Process control** | $O_2$ analyser upstream and downstream |
| **Common problems** | Catalyst poisoning, temperature rise if $O_2$ high |
| **Safety** | Exothermic reaction – high-$O_2$ trip |

## Step 7 – Hydrogen drying

| Aspect | Details |
|---|---|
| **Purpose** | Reduce moisture to the dew point required by the user (e.g. fuel-cell grade ISO 14687) |
| **Inputs** | Saturated $H_2$ |
| **Outputs** | Dry $H_2$ |
| **Equipment** | Temperature-swing adsorption (TSA) dryer with twin molecular-sieve beds, coolers, knock-out pot |
| **Operating parameters** | Dew point specification per end use |
| **Process control** | Dew-point analyser, automatic bed switching and regeneration |
| **Common problems** | Bed saturation, regeneration gas losses |
| **Safety** | Hot regeneration gas is hydrogen – leak tightness |

## Step 8 – Compression

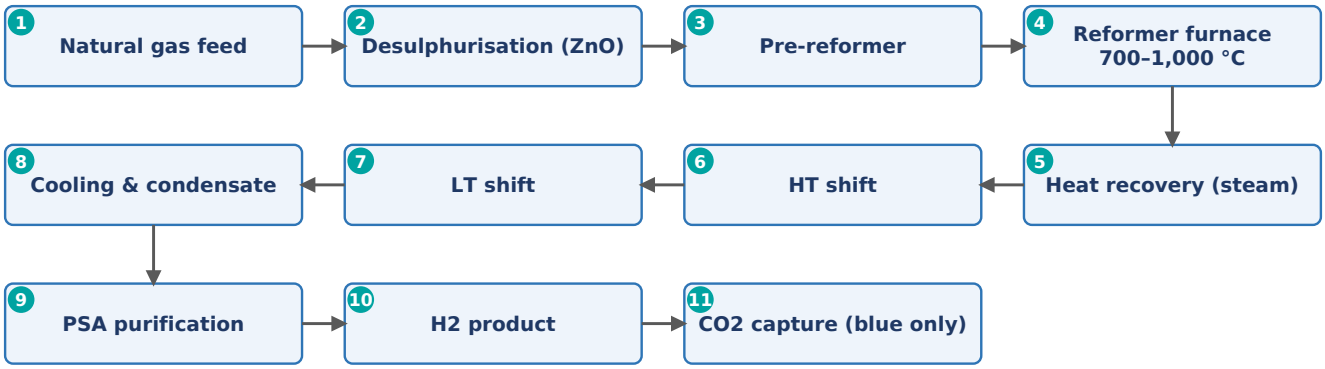
| Aspect | Details |
|---|---|
| **Purpose** | Raise pressure for storage, pipelines or vehicles |
| **Inputs** | Dry $H_2$ at electrolyser outlet pressure (atmospheric – 30 bar typical [R1]) |
| **Outputs** | $H_2$ at storage / delivery pressure |
| **Equipment** | Diaphragm, reciprocating piston or ionic compressors; intercoolers |
| **Operating parameters** | Multi-stage with intercooling; 70 bar reduces volume ~65×, 1,000 bar ~625× vs atmospheric [R1] |
| **Process control** | Suction/discharge pressure and temperature, vibration, diaphragm leak detection |
| **Common problems** | Valve and seal wear, diaphragm failure, oil contamination (lubricated machines) |
| **Safety** | High pressure + $H_2$; embrittlement-resistant materials; relief & ESD |

### Step 9 – Storage & dispatch

| Aspect | Details |
|---|---|
| **Purpose** | Buffer production vs. demand and deliver to customers |
| **Inputs** | Compressed $H_2$ |
| **Outputs** | $H_2$ to pipeline, tube trailers or end user |
| **Equipment** | Pressure vessels / cylinder banks, pipelines, dispensers, tube-trailer filling bays |
| **Operating parameters** | Storage pressure set by vessel rating (Module 8) |
| **Process control** | Inventory (pressure), flow metering, quality sampling |
| **Common problems** | Fatigue from pressure cycling |
| **Safety** | Separation distances, gas detection, fire protection |

## 3.2 Grey / blue hydrogen by SMR – process walk-through

**SMR chain (reformer conditions per US DOE [R2])**




| Step | Purpose | Key equipment | Watch-outs |
|---|---|---|---|
| **Desulphurisation** | Protect nickel catalyst from sulphur | Hydrogenator + ZnO beds | Bed breakthrough |
| **Reforming** | $CH_4 + H_2O \rightarrow CO + 3H_2$ | Catalyst tubes in fired furnace | Tube metal temperature, carbon formation (keep steam/carbon ratio) |
| **Heat recovery** | Use hot syngas & flue gas to make steam | Waste-heat boiler, steam drum | Boiler water quality |
| **Shift (HT/LT)** | $CO + H_2O \rightarrow CO_2 + H_2$ | Fixed-bed reactors | Catalyst activity, temperature control |
| **PSA** | Remove $CO_2$, CO, $CH_4$, $H_2O$ | Multi-bed adsorbers, valve skid | Valve reliability, recovery vs purity |
| **$CO_2$ capture** | Make blue hydrogen | Amine absorber/stripper, compression | Energy for regeneration, storage site |

# Quiz & revision – Module 3

Answer first, then check the key at the end of the module. Score yourself in the Progress Tracker (Appendix D).

**Q1.** Why is a de-oxo unit needed after a PEM/AEL stack?
**a)** To add oxygen **b)** To remove crossover O2 **c)** To remove CO2 **d)** To cool the gas

**Q2.** Typical raw water use per kg H2 including treatment losses (IRENA):
**a)** 2–4 kg **b)** 9 kg **c)** 18–24 kg **d)** 100 kg

**Q3.** Which dryer type uses twin molecular-sieve beds with heat regeneration?
**a)** PSA **b)** TSA **c)** Membrane **d)** Cyclone

**Q4.** Why must sulphur be removed before SMR?
**a)** It poisons the nickel catalyst **b)** It increases yield **c)** It is a product **d)** It reduces steam use

**Q5.** Compressing H2 from 1 to 70 bar reduces its volume by about:
**a)** 7× **b)** 65× **c)** 625× **d)** 870×

**Answer key**
**Q1 – b**: trace O2 is catalytically reacted to water; **Q2 – c**: [R1]; **Q3 – b**: temperature-swing adsorption; **Q4 – a**: ; **Q5 – b**: [R1]